

BIOGRAPHY

I am a postdoctoral researcher (Shuimu Scholar) at the Institute of Future Cities and Infrastructures, Tsinghua University, working with [Prof. Dongping Fang](#). I received my Ph.D. degree from the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, supervised by [Prof. Zhiguo Cao](#). My research interests lie in (1) 3D vision and image manipulation, with a particular emphasis on computational photography; (2) multimodal language models and their applications in construction automation.

EMPLOYMENT

Postdoctoral Researcher Institute of Future Cities and Infrastructures, Tsinghua University Advisor: Prof. Dongping Fang . I lead a project on multi-modal LLMs in construction safety.	Aug 2025 – Now
Project Officer S-Lab for Advanced Intelligence, Nanyang Technological University Advisor: Prof. Guosheng Lin . I lead a project on high-fidelity human avatar modeling.	Nov 2023 – Nov 2024

EDUCATION

Huazhong University of Science and Technology, China Doctor of Artificial Intelligence Thesis: "All-in-Focus Novel View Synthesis from Defocus-Blurred Images"	Sep 2020 – Jun 2025 GPA: 90.19
Huazhong University of Science and Technology, China Bachelor of Automation	Sep 2016 – Jun 2020 GPA: 89.2

PUBLICATIONS

Journal Articles

Defocus to Focus: Photo-realistic Bokeh Rendering by Fusing Defocus and Radiance Priors Xianrui Luo*, Juewen Peng*, Ke Xian, Zijin Wu, Zhiguo Cao (*Equal Contribution)	Information Fusion 2023 [Paper] [Code]
Point-and-Shoot All-in-Focus Photo Synthesis From Smartphone Camera Pair Xianrui Luo, Juewen Peng, Weiyue Zhao, Ke Xian, Hao Lu, Zhiguo Cao	IEEE TCSVT 2023 [Paper]
Dual-Camera All-in-Focus Neural Radiance Fields Xianrui Luo, Zijin Wu, Juewen Peng, Huiqiang Sun, Zhiguo Cao, Guosheng Lin	IEEE TPAMI 2025 [Paper] [Code]
BokehMe++: Harmonious Fusion of Classical and Neural Rendering for Versatile Bokeh Creation Juewen Peng, Zhiguo Cao, Xianrui Luo, Ke Xian, Wenfeng Tang, Jianming Zhang, Guosheng Lin	IEEE TPAMI 2024 [Paper]

Conference Papers

Dynamic Neural Radiance Field From Defocused Monocular Video Xianrui Luo, Huiqiang Sun, Juewen Peng, Zhiguo Cao	ECCV 2024 [Paper] [Code]
Bokeh Rendering from Defocus Estimation Xianrui Luo*, Juewen Peng*, Ke Xian, Zijin Wu, Zhiguo Cao (*Equal Contribution)	ECCVW 2020 [Paper]
BokehFlow: Depth-Free Controllable Bokeh Rendering via Flow Matching Yachuan Huang, Xianrui Luo, Qiwen Wang, Liao Shen, Jiaqi Li, Huiqiang Sun, Zihao Huang, Wei Jiang, Zhiguo Cao	AAAI 2026 [Paper]
Interactive Portrait Bokeh Rendering System Juewen Peng, Xianrui Luo, Ke Xian, Zhiguo Cao	ICIP 2021 [Paper]
BokehMe: When Neural Rendering Meets Classical Rendering Juewen Peng, Zhiguo Cao, Xianrui Luo, Hao Lu, Ke Xian, Jianming Zhang	CVPR 2022 [Paper] [Code]
MPIB: An MPI-Based Bokeh Rendering Framework for Realistic Partial Occlusion Effects Juewen Peng, Jianming Zhang, Xianrui Luo, Hao Lu, Ke Xian, Zhiguo Cao	ECCV 2022 [Paper] [Code]
Fast Full-frame Video Stabilization with Iterative Optimization Weiyue Zhao, Xin Li, Zhan Peng, Xianrui Luo, Xinyi Ye, Hao Lu, Zhiguo Cao	ICCV 2023 [Paper] [Code]

Selective Bokeh Effect Transformation

Juewen Peng, Zhiyu Pan, Chengxin Liu, **Xianrui Luo**, Huiqiang Sun, Liao Shen, Ke Xian, Zhiguo Cao

CVPRW 2023

[\[Paper\]](#)[\[Code\]](#)

BokehCrafter: Taming Video Diffusion Models for Controllable Bokeh Rendering

AAAI 2026

Qiwen Wang, Liao Shen, Jiaqi Li, Tianqi Liu, Huiqiang Sun,
Zihao Huang, Yachuan Huang, **Xianrui Luo**, Zhiguo Cao

PROJECT

Shenzhen Construction Engineering Group Funding Project on Safety Inspection

Sep 2025 – Now

- Develop a reinforcement learning-based fine-tuning framework for pretrained vision-language models to improve automated construction safety hazard inspection under image-level weak supervision.
- Reformulate safety inspection from direct hazard classification to evidence-based visual reasoning, enabling the model to localize hazard-relevant visual evidence before generating final safety judgments.
- Curate and quality-control real-world construction safety inspection data, including label cleaning, duplicate filtering, and benchmark construction for evaluating domain-adapted VLMs.

S-Lab Funding Project on Human Avatar Modeling

Nov 2023 – Nov 2024

- Build and clean large-scale motion datasets for character animation, improving data quality and reliability for downstream action generation model training.
- Develop data capture and synthesis pipelines for high-fidelity human avatar modeling, supporting research on 3D human reconstruction and photorealistic avatar generation.

NTIRE 2023 Challenge on Bokeh Effect Transformation (CVPR2023 Workshop)

Jun 2023

Winner Award

Vivo Funding Project on Simulation of Bokeh Effect from DSLR

Jan 2023 – Dec 2023

I serve as a core member responsible for realistic bokeh synthesis for smartphone photography, I analyze experimental results and provide ideas to improve visualizations.

AIM 2020 Challenge on Rendering Realistic Bokeh (ECCV2020 Workshop)

Aug 2020

Runner Up Award

AWARDS AND HONORS

National Scholarship (1% of all students)	<i>Huazhong University of Science and Technology</i>	2024
Academic Scholarship	<i>Huazhong University of Science and Technology</i>	2020 – 2024
Outstanding Graduates	<i>Huazhong University of Science and Technology</i>	2020
Merit Student (5% of all students)	<i>Huazhong University of Science and Technology</i>	2016, 2023
Best Poster Award	<i>Artificial Intelligence Conference and Entrepreneurs Summit Forum of China's Optics Valley</i>	2022

SKILLS

Programming Language

Python, MATLAB, C

Machine Learning Tools

PyTorch, OpenCV

English

TOEFL 109, GRE 325 (AW 4.0)